

Report V

Diagonalization of matrix with power method and Gram-Schmidt orthogonalization

Author: Patryk Kuna

1. Introduction

In many physical problems, especially those involving differential equations, the second derivative operator plays a fundamental role. When discretized using the finite difference method, it leads to a matrix eigenvalue problem.

In this project, we consider the approximation:

$$\frac{d^2 f_i}{dx^2} \approx \frac{f_{i+1} - 2f_i + f_{i-1}}{\Delta^2}$$

which results in a symmetric tridiagonal matrix. The matrix elements are defined as:

$$a_{i,j} = \begin{cases} -2 & \text{for } j = i \\ 1 & \text{for } j = i \pm 1 \\ 0 & \text{otherwise} \end{cases}$$

The goal of this exercise is to determine eigenvalues and eigenvectors of this matrix numerically.

To solve the problem, the **power method** was used. This iterative algorithm allows approximation of eigenvalues by repeated multiplication of a vector by the matrix. In its basic version, it converges to the eigenvalue of the largest magnitude.

To obtain all eigenvalues, the method is extended using the **Gram-Schmidt orthogonalization**, which removes contributions from already computed eigenvectors, allowing convergence to subsequent eigenpairs.

2. Description of the Numerical Task

The matrix was constructed for:

- $n = 6$
- $it_{max} = 150$
- $\varepsilon = 10^{-8}$

The task consisted of:

- constructing the matrix using finite difference approximation
- verifying correctness by printing the matrix
- iteratively computing eigenvalues and eigenvectors
- saving eigenvalue approximations during iterations
- reconstructing the matrix and its inverse

The reconstruction formulas are:

$$A_{num} = \sum_{k=0}^{n-1} \lambda_k \vec{x}_k \vec{x}_k^T$$

$$A_{num}^{-1} = \sum_{k=0}^{n-1} \lambda_k^{-1} \vec{x}_k \vec{x}_k^T$$

3. Numerical Method

The algorithm followed the improved power method described in the instruction file.

For each eigenvalue:

- a random initial vector is generated
- matrix-vector multiplication is performed
- eigenvalue is estimated using the formula:

$$\lambda = \frac{\vec{x}^T A \vec{x}}{\vec{x}^T \vec{x}}$$

Then:

- Gram-Schmidt orthogonalization is performed

$$\vec{y} \leftarrow \vec{y} - \sum_{m=0}^{k-1} (\vec{x}_m^T \vec{y}) \vec{x}_m$$

- the vector is normalized
- convergence is checked using relative error

The process is repeated until the stopping condition is satisfied.

4. Implementation

The program was written in C++ and does not use any external numerical libraries, the scalar product is calculated iteratively.

The main steps of the implementation are:

1. Definition of parameters:
 - $n = 6, it_{max} = 150, \varepsilon = 10^{-8}$
2. Construction of the matrix:
 - filling diagonal and off-diagonal elements according to finite difference scheme

$$A = \begin{bmatrix} -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 1 & -2 \end{bmatrix}$$

3. Implementation of basic operations:

- scalar product
- matrix-vector multiplication
- vector normalization

4. Implementation of Gram-Schmidt orthogonalization:
 - removal of projections on previously computed eigenvectors
5. Iterative computation:
 - calculation of eigenvalues and eigenvectors
 - saving intermediate results to files
6. Reconstruction of matrices:
 - computation of A_{num}
 - computation of A_{num}^{-1}
7. Saving results:
 - eigenvalues and eigenvectors
 - reconstructed matrices

5. Results

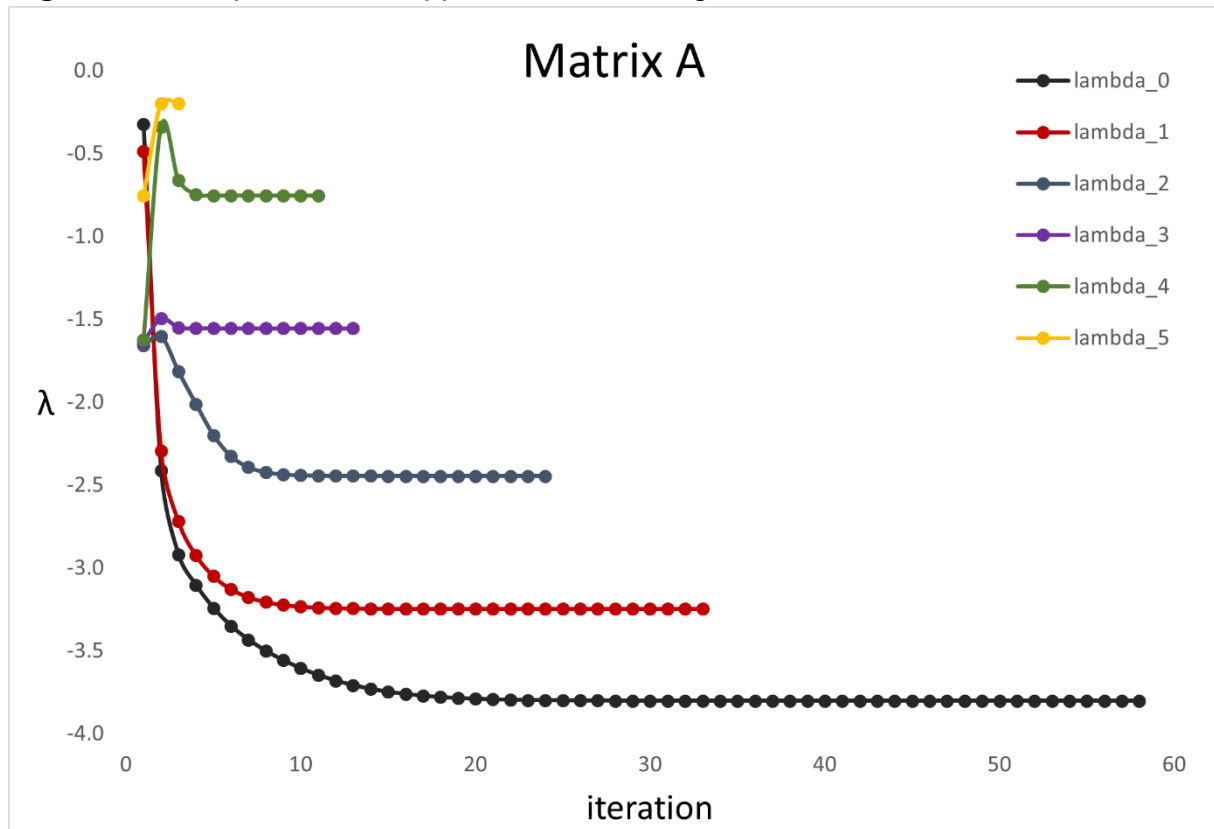
Eigenvalue convergence

The plot (Fig. 1) shows the convergence of eigenvalues as a function of iteration number.

All eigenvalues converge smoothly to some stable values.

- the first eigenvalue converges the fastest
- higher-index eigenvalues require more iterations
- convergence is reached well before it_{max}

Figure 1. Example iterative approximations of eigenvalues for matrix A



Final eigenvalues

The obtained eigenvalues are approximately:

$$\lambda_0 \approx -3.80, \lambda_1 \approx -3.25, \lambda_2 \approx -2.45, \lambda_3 \approx -1.56, \lambda_4 \approx -0.75, \lambda_5 \approx -0.20$$

They follow the expected distribution for a discrete second derivative operator.

Matrix reconstruction

The reconstructed matrix A_{num} is consistent with the original matrix.

- very good agreement between elements
- numerical errors are negligible

Inverse matrix

The inverse matrix is symmetric - this kind of behavior is expected because, it is constructed from the symmetric components $\vec{x}_k \vec{x}_k^T$

6. Conclusions

Numerical eigenvalues are not exact, but they approximate the true values very well. The convergence is fastest for eigenvalues with the largest magnitude, while higher-index eigenvalues require more iterations. This behavior is characteristic of the power method.

The reconstruction of the matrix confirms correctness of the obtained eigenpairs. The inverse matrix is symmetric, which follows from the symmetry of the original matrix and orthogonality of eigenvectors.

It was also observed that Gram-Schmidt orthogonalization is necessary. Without it, the method would converge repeatedly to the same eigenvector and it would not be possible to determine all eigenvalues.

The number of iterations required for convergence increases with the eigenvalue index, which is expected since the power method first captures dominant components.

Using a less restrictive stopping criterion (e.g. $\varepsilon = 10^{-4}$) leads to a faster convergence but lower accuracy of the reconstructed matrix.